

CREDIT CARD APPROVAL PREDICTION USING MACHINE LEARNING

1. Project Title

Credit Card Approval Prediction Using Machine Learning

2. Abstract

The Credit Card Approval Prediction project is a Machine Learning-based web application developed to predict whether a customer's credit card application will be approved or rejected. The project uses applicant details such as income, age, employment, education, family status, housing type, and occupation to make predictions. The data is preprocessed using feature engineering and label encoding, and the XGBoost algorithm is used to build the prediction model. A Flask web application provides an interactive interface where users can enter applicant details and instantly receive the prediction result.

3. Problem Statement

Banks receive thousands of credit card applications every day. Evaluating every application manually is time-consuming and may lead to inconsistent decisions. The objective of this project is to automate the credit card approval process using Machine Learning, enabling faster and more accurate prediction of application approval based on customer information.

4. Objectives

- Develop a Machine Learning model for predicting credit card approval.
- Perform data preprocessing and feature engineering.
- Compare multiple Machine Learning algorithms.
- Select the best-performing model.
- Deploy the model using Flask.
- Provide a simple web interface for prediction.

5. Dataset

Dataset Used:

- ❖ Application Record Dataset
- ❖ Credit Record Dataset

Features Used:

- ❖ Gender
- ❖ Car Ownership
- ❖ Property Ownership

- ❖ Annual Income
- ❖ Income Type
- ❖ Education Level
- ❖ Family Status
- ❖ Housing Type
- ❖ Occupation Type
- ❖ Family Members
- ❖ Age
- ❖ Years Employed

6. Technologies Used

Technology	Purpose
Python	Programming Language
Jupyter Notebook	Model Development
Pandas	Data Preprocessing
NumPy	Numerical Operations
Scikit-learn	Data Splitting, Scaling & Evaluation
XGBoost	Machine Learning Algorithm
Flask	Web Application Framework
HTML	User Interface
Pickle	Saving Model and Encoders
VS Code	Flask Development

7. Project Methodology

Step 1

Data Collection

The Application Record and Credit Record datasets were collected and merged.

Step 2

Data Preprocessing

- Removed unnecessary columns
- Handled missing values
- Converted DAYS_BIRTH into AGE
- Converted DAYS_EMPLOYED into YEARS_EMPLOYED
- Label encoded categorical variables
- Standardized numerical features

Step 3

Model Development

The following Machine Learning models were trained:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

Step 4

Model Evaluation

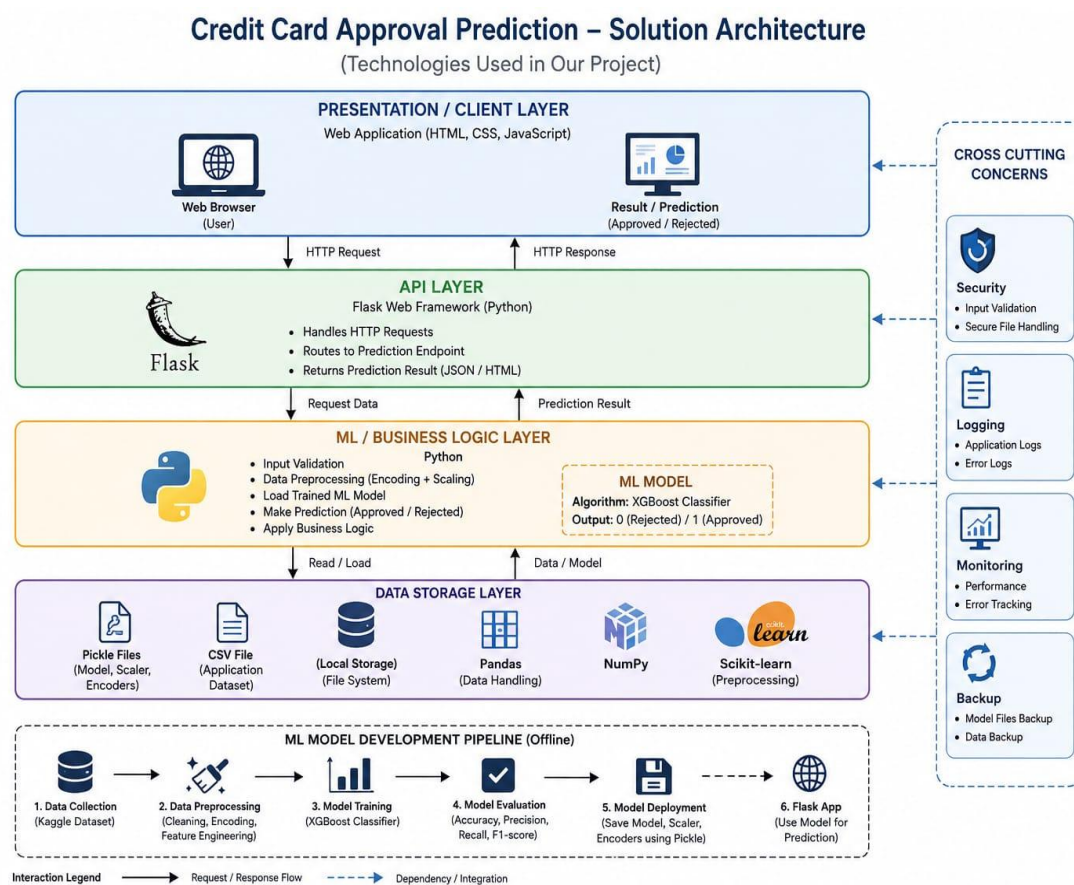
The models were compared based on prediction accuracy.

Step 5

Deployment

The best-performing XGBoost model was saved using Pickle and deployed through a Flask web application.

8. Project Architecture



9. Machine Learning Models Compared

Model	Purpose
Logistic Regression	Baseline Classification Model
Decision Tree	Tree-based Classification
Random Forest	Ensemble Learning
XGBoost	Final Selected Model

- The XGBoost model achieved the highest performance and was selected for deployment.

10. Model Performance

Accuracy : 98.41%

Evaluation Metrics:

- Accuracy
- Confusion Matrix
- Classification Report

XGBoost Report

Accuracy: 0.984092155787164

Confusion Matrix:

```
[[7165  10]
 [ 106  11]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	7175
1	0.52	0.09	0.16	117
accuracy			0.98	7292
macro avg	0.75	0.55	0.58	7292
weighted avg	0.98	0.98	0.98	7292

11. Flask Web Application

The trained model was integrated into a Flask web application.

Users can:

- Enter applicant information
- Click the Predict button
- Receive Credit Card Approved or Credit Card Rejected as output

Credit Card Approval Prediction

Fill in the applicant details to predict credit card approval.

Gender	Own Car
Select Gender	Select
Own Realty	Annual Income
Select	Enter Annual Income
Income Type	Education
Select Income Type	Select Education
Family Status	Housing Type
Select Family Status	Select Housing Type
Occupation	Number of Family Members
Select Occupation	Enter Number of Family Members
Age	
Enter Age	
Years Employed	
Enter Years Employed	

Predict Credit Card Approval

Credit Card Approval Prediction

Fill in the applicant details to predict credit card approval.

Gender	Own Car
Female	No
Own Realty	Annual Income
Yes	216000
Income Type	Education
Pensioner	Secondary / Secondary Special
Family Status	Housing Type
Widow	House / Apartment
Occupation	Number of Family Members
Unknown	1
Age	
54	
Years Employed	
4	

Predict Credit Card Approval

Credit Card Approved 

12. Features

- Data preprocessing
- Feature engineering
- Label encoding
- Feature scaling
- Multiple ML model comparison
- XGBoost implementation
- Flask deployment
- Web-based prediction interface

13. Future Scope

- Improve prediction accuracy using additional datasets.

- Deploy the application to cloud platforms such as Render or Railway.
- Integrate with a real-time banking database.
- Add user authentication.
- Develop a mobile application.

14. Conclusion

The Credit Card Approval Prediction project successfully demonstrates the application of Machine Learning in automating credit card approval decisions. By comparing multiple classification algorithms, XGBoost was selected as the final model due to its superior performance. The Flask-based web application provides an easy-to-use interface for predicting approval status, making the system practical, efficient, and scalable.